**Summary Post**

My initial post discussed the issues caused by a server failure within my company which provides accounting software. This failure was compounded by lack of monitoring. My initial feeling was that the speed of the changes that have come about as part of the 4th Industrial Revolution have not been matched by a speed of governance changes.

Reading and replying to other posts and the responses to my post have brought about the understanding that industry 5.0 has emerged, in part, to address exactly this issue.

However, how much resilience can be built in to systems is questionable. Human beings can not match the processing speed or connectivity of today's technology, so failures will could potentially have a greater impact on businesses than they ever could before.

Both John-Jon Steyn and Ali Nahas made a very interesting point around governance of outsourced software, pointing out that the business still retains responsibility for the accessibility and backups of data, even when the hosting of this is outsourced. Ababneh et al (2026) and NCSC (2022) both support these comments.

Emily Shaibu also gave a very considered response to my initial post. Emily suggested that the views of Metcalf (2024), focusing on human-centric resilience are important and advised that Xu et al (2021) covered my points about ensuring that the speed of innovation in the 4th Industrial Revolution is met with equally swift development of monitoring and governance.

In conclusion, my initial post engendered some really thoughtful responses from my peers, for which I'm extremely grateful. I believe the general consensus across all the articles I've read, and the thoughts of my peers, is that the ability to monitor and govern today's technology will always be slightly behind the technology itself. Whilst technology increases in speed and connectivity at a record pace, the ability to build resilience checks and governance into these breakthroughs has to remain a step behind, by its very nature.

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**Initial Post**

I work in accounting software. Recently, one of my company's servers failed. The server monitoring failed to pick it up because the monitoring software was not running. Therefore multiple customers – large and small businesses - were unable to access the software for a short time. Following reports of the outage from our customers, situation was quickly resolved. The time taken for our reputation to recover with those customers was much longer. The internal investigations led to an improved "playbook" to ensure there could be no repeat of this event.

In my working lifetime, I've moved from recording accounting transactions on paper, to recording them on a standalone PC, to entering them on a cloud based platform. The reliance for backups and accessibility now lies with an outside contractor. The swiftness of this change – it's taken around 20 years for small companies to leave behind centuries of using hand-written ledgers, to them being able to access cloud based, multi-featured solutions - demonstrates that this is the 4th Industrial revolution due to its "velocity, scope, and systems impact". (Schwab 2016)

Both Schwab (2016) and Metcalf (2024) speak about the challenges faced by governments to keep pace with regulating new technologies, but I believe this is also true for internal governance in companies using new technologies.

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**Response 1**

by Ali Nahhas - Monday, 10 August 2026, 6:37 PM

I agree that internal governance can struggle to keep pace with technology, but I think this incident points to something specific. The problem was not that infrastructure moved to the cloud. It was that the organisation relied on a monitoring system that could itself fail silently. Cotroneo et al. (2023) show that silent failures are difficult in cloud environments because conventional logging may not generate a warning. In that sense, customer reports became the real monitoring system, which is a weak position for any accounting platform where availability affects business operations.

I would therefore question whether an improved playbook is enough. Mushtaq et al. (2024) distinguish reactive fault handling from proactive and resilient approaches, where systems continuously monitor faults and continue operating when components fail. Applying that idea here could mean independent health checks, redundant monitoring, automated failover and alerts confirming the monitoring service itself remains active.

There is also a governance issue around outsourcing. Ababneh et al. (2026) argue that cloud security follows a shared-responsibility model. Moving infrastructure to an external provider therefore changes responsibility rather than removing it. For me, the Industry 5.0 lesson is designing accountability and resilience into the system before customers become the outage alarm.

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**Response 2**

by John-Jon Steyn - Sunday, 16 August 2026, 3:32 PM

Hey Sarah,

Your example makes the governance problem tangible because the primary server fault was compounded by failure of the control intended to detect it. I agree that the resulting trust damage matters as much as the short interruption: for accounting software, availability affects payroll, invoicing and reporting deadlines, so even a brief outage can create consequences beyond the provider.

A useful extension is to distinguish monitoring the infrastructure from testing the service as a customer experiences it. Google's site-reliability guidance separates internal telemetry from external checks of observable behaviour (Ewaschuk, 2016). In your case, an independent synthetic login or transaction, run outside the affected environment, could have alerted the team even when the monitoring agent stopped. The design should also generate an alert when an expected monitoring signal disappears.

I would slightly qualify the idea that responsibility for accessibility and backups now lies with the contractor. The NCSC (2022) explains that cloud providers and customers retain different responsibilities: infrastructure may be delegated, but the customer must still determine whether the service meets its requirements and understand which controls it retains. Contracted recovery objectives, backup testing, audit evidence and an exit route therefore remain internal governance concerns.

The improved playbook is a strong response, provided it is exercised rather than only documented. NIST contingency-planning guidance links effective recovery to assigned responsibilities, testing and regular maintenance (Swanson et al., 2010). I would include ownership, escalation, an independent status channel, recovery priorities and customer communications. This makes Industry 5.0 concrete: resilience protects customers through verified controls and clear human responsibility, rather than assuming that cloud automation is self-supervising (Metcalf, 2024).

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by Emily Shaibu - Wednesday, 19 August 2026, 4:22 PM
Hello Sarah,

Thank you for sharing this candid reflection on a real-world systems failure. Your example powerfully illustrates how even seemingly minor technical oversights can have disproportionate reputational consequences, and you observation about the velocity of change outpacing internal governance is particularly astute.

The failure you describe - where the monitoring software itself was not running - represents a classic single point of failure. Several preventative measures could have mitigated this risk. Firstly, implementing redundant monitoring systems would address this directly, A second day, independent monitoring solution - perhaps cloud-based and separate from the primary infrastructure- could have alerted your team when the primary monitoring software ceased functioning (Abomhara, 2015). This aligns with the resilience-by-design principle central to Industry 5.0's vision (Xu et al., 2021).

Secondly, a heartbeat or watchdog mechanism could have been implemented. This involves a simple, automated process that periodically checks whether the monitoring software is operational and sends alerts via an alternative channel, such as SMS or a third-party notification service, if it detects failure. This is a low-cost but highly effective safeguard.

Thirdly, your improved playbook is commendable, but it would benefit from regular, scheduled drills that simulate monitoring failure scenarios. Such exercises would ensure that staff remain familiar with manual detection and escalation procedures, transforming theoretical knowledge into practical readiness. This human-centric approach to resilience is precisely what Metcalf (2024) advocates when arguing that socio-technical systems must integrate human capability with technological infrastructure.

Your point about internal governance struggling to keep pace with technological change resonates strongly. Companies often adopt cloud solutions rapidly without commensurately updating their risk management frameworks. This suggests that Industry 5.0's emphasis on human-centricity should extend to governance structures, ensuring that technological adoption is accompanied by equally agile oversight mechanisms (Xu et al., 2021)

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